STRUCTURAL LOAD ANALYSIS REPORT

Project Element:	Single-Span Reinforced Concrete Floor Beam (Beam B1)
Structure Type:	Multi-Story Residential Building Framework
Analysis Methodology:	Equivalent Static Method & Limit State Design (Ultimate Limit State)
Date:	May 18, 2026

1. Introduction & Structural Parameters

This engineering report presents the comprehensive load evaluation and internal structural behavior analysis for a representative interior, simply supported reinforced concrete (RC) beam. The analysis ensures compliance with safety requirements under ultimate design conditions, quantifying the vertical gravity impacts alongside basic lateral seismic coefficients.

1.1 Material & Geometric Framework

- ***Beam Span (L)***: 6.0 m (Center-to-center distance between supports)
- ***Beam Section Dimensions***: Width $b = 300\ mm\ (0.3\ m)$, Total Depth $D = 500\ mm\ (0.5\ m)$
- ***Slab Thickness*** (t_s): 120 mm (0.12 m)
- ***Tributary Width*** (W_t): 3.5 m (Floor slab surface clearing directly transferring load into the beam)
- ***Reinforced Concrete Density*** (γ_c): 25 kN/m³
- ***Floor Finish Density*** (γ_{ff}): 24 kN/m³

2. Load Estimation & Distribution

Loads are categorized structurally based on execution duration: dead loads (permanent static weights) and live loads (transient variable occupancy forces).

2.1 Dead Load (DL) Calculation

Dead loads are calculated directly from component volume bounds and material densities:

1. Beam Self-Weight: $w_{beam} = b \times D \times \gamma_c = 0.3 \times 0.5 \times 25 = 3.75\ kN/m$
2. Slab Load Contribution: $w_{slab} = t_s \times W_t \times \gamma_c = 0.12 \times 3.5 \times 25 = 10.50\ kN/m$
3. Architectural Floor Finishes (50mm depth assumed): $w_{ff} = 0.05 \times 3.5 \times 24 = 4.20\ kN/m$

Summing these components yields the total continuous dead load acting on the beam line:

$$w_{DL} = 3.75 + 10.50 + 4.20 = 18.45\ kN/m$$

2.2 Live Load (LL) Calculation

According to standard occupancy classifications for residential space, an un-factored design live load threshold of 2.0 kN/m² is mapped to the beam:

$$w_{LL} = Intensity \times W_t = 2.0\ kN/m^2 \times 3.5\ m = 7.00\ kN/m$$

2.3 Lateral Earthquake Load (EL) Evaluation

Using a simplified Equivalent Static Method for low-to-moderate seismic boundaries, the operational seismic floor weight (W) utilizes full dead load combined with a 25% live load reduction coefficient:

$$W = (w_{DL} + 0.25 \cdot w_{LL}) \times L = (18.45 + 0.25 \times 7.00) \times 6.0 = 121.20\ kN$$

Applying a horizontal seismic base coefficient ($A_h = 0.05$):

$$V_b = A_h \times W = 0.05 \times 121.20 = 6.06\ kN$$

This horizontal shear shifts into the structural framing lines as an induced axial force and localized frame end moments, modeled conservatively for this single element as a reversible support moment effect: $M_{EL} = \pm 9.0\ kNm$.

3. Design Load Combinations (Ultimate Limit State)

Safety factors scale nominal values to generate ultimate cross-sectional demand profiles:

- **Combination 1 (Pure Gravity Ultimate):**
  $w_u = 1.5 \cdot w_{DL} + 1.5 \cdot w_{LL} = 1.5(18.45) + 1.5(7.00) = 38.18\ kN/m$
- **Combination 2 (Seismic Structural Boundary):**
  $w_{u,eq} = 1.2 \cdot w_{DL} + 1.2 \cdot w_{LL} = 1.2(18.45) + 1.2(7.00) = 30.54\ kN/m$
  $M_{u,eq} = 1.2 \cdot M_{EL} = 1.2 \times 9.0 = \pm 10.80\ kNm$

4. Structural Behavior & Internal Forces Analysis

Under a uniformly distributed load, a simply supported span generates standard internal stress responses: shear force tracks linearly from supports down to zero at center-span, while bending moment describes a parabolic path peaking at mid-span.

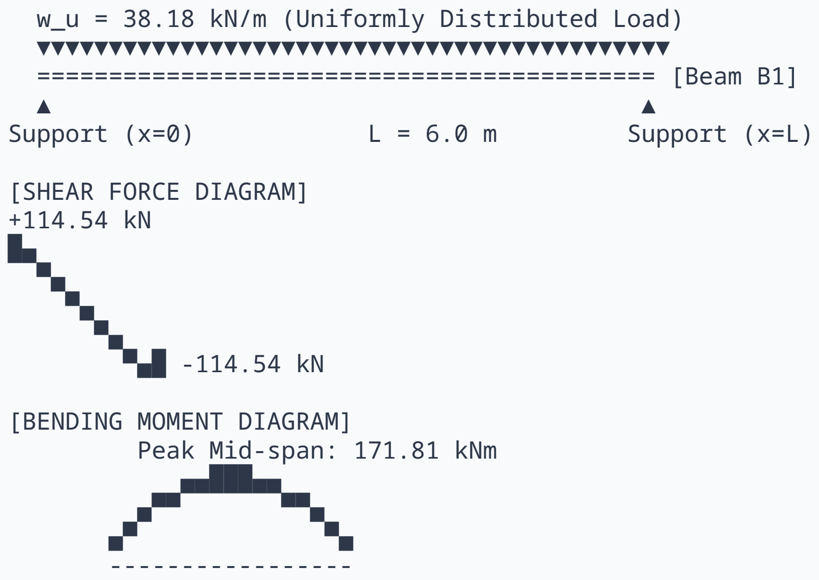

Figure 1: Load Distribution, Shear Force Diagram (SFD), and Bending Moment Diagram (BMD)

4.1 Maximum Design Shear Force (V_u)

Peak shear stress aggregates at the face of the supports ($x = 0$ and $x = 6.0\ m$) governed by Combination 1:

$$V_u = (w_u \cdot L) / 2 = (38.18 \times 6.0) / 2 = 114.54\ kN$$

4.2 Maximum Design Bending Moment (M_u)

Maximum sagging flexural moment locates exactly at mid-span ($x = L/2$):

$$M_u = (w_u \cdot L^2) / 8 = (38.18 \times 6.0^2) / 8 = 171.81\ kNm$$

Evaluating Combination 2 (Seismic flexural combination) at center-line:

$$M_{mid,eq} = (w_{u,eq} \cdot L^2) / 8 + M_{u,eq} = (30.54 \times 36) / 8 + 10.80 = 148.23\ kNm$$

Comparing outcomes, **Combination 1 (1.5DL + 1.5LL) governs** the ultimate limits of the beam section profile.

5. Structural Calculation Sheet (Summary Table)

Load Type / Action Source	Unfactored Value	Distribution Framework	Governing Design Demand
Dead Load: Beam Self-Weight	3.75 kN/m	Linear cross-section profile calculation	**w_{DL} = 18.45 kN/m**
Dead Load: Slab Load Distribution	10.50 kN/m	Tributary floor area strip allocation	
Dead Load: Architectural Floor Finishes	4.20 kN/m	Uniform floor screed/tile depth allocation	
Live Load: Occupancy Load Profile	7.00 kN/m	Residential variable building code specification	**w_{LL} = 7.00 kN/m**
Earthquake Load: Lateral Base Shear	6.06 kN	Equivalent lateral mass translation matrix	$M_{EL} = \pm 9.00$ kNm
Ultimate Shear Force (V_u)	—	Critical section at support inner face	**114.54 kN**
Ultimate Bending Moment (M_u)	—	Critical section at beam mid-span point	**171.81 kNm**

Engineering Summary Note: Shear design must employ vertical stirrups closely spaced near support vertices, tapering down safely toward mid-span. Flexural design requires longitudinal steel areas sufficient to accommodate 171.81 kNm within standard strain limits.